

MATH 110C, HOMEWORK 1

DUE TUESDAY APRIL 16

Exercise 1 Let K/F be a field extension. Assume that the characteristic of F is not 2 and that $[K : F] = 2$. Show that there exists $\alpha \in K$ such that $\alpha^2 \in F$ and $K = F(\alpha)$.

Exercise 2 Let K/F be a field extension and let $\alpha, \beta \in K$. Assume α and β are algebraic over F , of respective degrees m and n .

(1) Let m' be the degree of α over $F(\beta)$. Show that β has degree $\frac{m'n}{m}$ over $F(\alpha)$.

(2) If m and n are coprime, show that $[F(\alpha, \beta) : F] = mn$.

Exercise 3 Consider two field extensions M/L and L/K . Let $\alpha \in M$ be an element that is algebraic over K with degree n over K and degree n' over L . Is it true that $n'|n$? If yes, give a proof. If no, give an example where it fails.

Exercise 4 Do Problems 2, 4, 7, 8 and 12 in Exercises 48.25, pp.297–298 of the book.